J3
J1
J2
VDD
GND
SWDIO
SWCLK
SWO
NRST
ST-LINK_SWDIO
ST-LINK_SWCLK
ST-LINK_SWO
IR_NRST
ST-LINK_UART1_RX
ST-LINK_UART1_TX
ST-LINK-IR
PINHD-1X8
ST-LINK_SWCLK
GY_RESET
S4
C17
1u
R4
2.2k
R3
2.2k
C13
100n
C14
10u
VIN
GND
SDA
SCL
VOUT
GND0
INT
RESET
+5V
C16
47u
C15
47u
JP1
DRB2
DRB1
KICKERYOBI
KICK1
BALL2
CURRENT_SENS_4
CURRENT_SENS_3
CURRENT_SENS_1
CURRENT_SENS_2
DRB_CS
COMMODULE_RX
COMMODULE_TX
INT
BALL1
JP3
TIM1_CH1
TIM1_CH3
BAT_SENS
UART5_RX
TIM1_CH4
KICK1
TIM1_CH1N
UART4_TX
TIM8_CH2N
TIM8_CH3
TIM1_CH2N
MD_RESET
TIM3_CH2
UART5_TX
TIM3_CH1
KICK2
UART4_RX
TIM1_CH3N
BAT_SENS
R6
1k
D2
C31
100n
D1
OPEN-MV
3.3V
VIN
SWC
SWD
RESET
BOOT
GND
SSEL
SCLK
MISO
MOSI
PWM1
PWM2
TXD/SCL
RXD/SDA
ADC/DAC
OPENMV3
UART2_RX
UART2_TX
>VALUES
TOUCH1_GPIO1_A0_D0
TOUCH2_GPIO2_A1_D1
TOUCH3_GPIO3_A2_D2
TOUCH4_GPIO4_A3_D3
TOUCH5_GPIO5_SDA_A4_D4
TOUCH6_GPIO6_SCL_A5_D5
GPIO43_TX_D6
TOUCH9_GPIO9_MOSI_A10_D10
TOUCH8_GPIO8_MISO_A9_D9
TOUCH7_GPIO7_SCK_A8_D8
GPIO44_RX_D7
5V
GND
3.3V
ESP32
ESP_LED
UART4_RX
UART4_TX
LED3
R14
330
SW2
R7
510
R21
4.7k
Q2
2SC2712-Y,LF
R10
10k
C19
100n
SW2_LED
SW3
R9
510
R8
4.7k
Q3
2SC2712-Y,LF
R11
10k
C18
100n
SW3_LED
VCC
SDA
SCL
GND
SW1
GOAL_SWITCH
R13
10k
LED2
R5
330
LED1
R2
330
U1
R1
10k
BOOT0
STM32F446RET6
PA0-WKUP(PA0)/TIM2_CH1/TIM2_ETR/TIM5_CH1/TIM8_ETR/USART2_CTS/UART4_TX/EVENTOUT
PA1/TIM2_CH2/TIM5_CH2/USART2_RTS/UART4_RX/QUADSPI_BK1_IO3/SAI2_MCLK_B/EVENTOUT
PA2/TIM2_CH3/TIM5_CH3/TIM9_CH1/USART2_TX/SAI2_SCK_B/EVENTOUT
PA3/TIM2_CH4/TIM5_CH4/TIM9_CH2/SAI1_FS_A/USART2_RX/OTG_HS_ULPI_D0/EVENTOUT
VDD
VDDA/VREF+
VBAT
VCAP_1
VSSA/VREF-
VSS
NRST
PC14-OSC32_IN(PC14)/EVENTOUT
PC15-OSC32_OUT(PC15)/EVENTOUT
PH0-OSC_IN/EVENTOUT
PH1-OSC_OUT/EVENTOUT
C1
C2
C3
C4
C5
C6
C7
C8
C9
4.7u
100n
1u
C10
S1
C12
15p
XTAL1
C11
15p
DB57
SS20ADFL
R12
300
SPKR
Q1
2SC2712-Y,LF
U2
UGCT7525AN4
POLARITY_PAD_(+)
POLARITY_PAD_(-)
DUMMY_PAD_2
DUMMY_PAD_1
DRB_CS
R20
1k
C30
100n
C20
100n
NRST2
S2
R15
10k
U3
STM32F303K8T6
BOOT0
NRST
VDD_1
VDD_2
VDDA/VREF+
VSS_1
VSS_2
PF0/OSC_IN
PF1/OSC_OUT
C21
C22
C24
C23
100n
4.7u
C25
1u
CURRENT_SENS_1
R16
1k
C26
100n
CURRENT_SENS_1IN
CURRENT_SENS_2
R17
C27
CURRENT_SENS_2IN
CURRENT_SENS_3
R18
C28
CURRENT_SENS_3IN
CURRENT_SENS_4
R19
C29
CURRENT_SENS_4IN
U4
U5
U6
VCC
TRIG
ECHO
GND
RCW-0001
TRIG_1
ECHO_1
TRIG_2
ECHO_2
TRIG_3
ECHO_3